THE UNIVERSITY OF BRITISH COLUMBIA

MICB 323 – MOLECULAR IMMUNOLOGY & VIROLOGY LABORATORY

COURSE LOGISTICS

WELCOME TO MICB 323

We look forward to working with all of you this term to do some science! This lab course is a structured inquiry lab course. This means we will provide central research questions, and guidance on the protocols to answer them. You will conduct experiments in the lab individually or with your peers; you will collect data and tell a story in the form of a scientific manuscript. You will gain fundamental technical skills in cell biological techniques, build analytical skills, evaluate experimental data, and practice scientific writing.

LECTURES: Note that class sessions will require active class engagement and will not be recorded.
Monday 1-3pm – P. A. Woodward Instructional Resources Centre (IRC) | Floor: G | Room: 1

LABS: All labs will take place in Room 3145 Biological Sciences, 1-5pm. The first lab will begin the week of Jan 12th.
L01 – Tuesday (Kabir Bhalla)
L02 – Wednesday (Sreeparna Vappala)
L03 – Thursday (Kabir Bhalla)
L04 – Friday (Sreeparna Vappala)

Teaching Team

Dr. Sreeparna Vappala (Course coordinator and Instructor, she/her)

- You can call me Sree.
- Email: s.vappala@ubc.ca
- Office: Biological Sciences, 3106
- I don't have set office hours for MICB 323, but I am always happy to meet. Reach out via email for additional support and we will set a mutually convenient time to meet either in person, in the lab or via zoom.

Kabir Bhalla (Instructor - he/him)

- Email: kabir.bhalla@msl.ubc.ca
- I don't have formal learning support hours for MICB 323, but I am very happy to meet whenever you need. Send me an email, and we will set up an in-person or virtual meeting.

TEACHING ASSISTANTS: connect with your TAs in the lab

- Cynthia Chung cynthia.chung@ubc.ca
- Josepha Klas jmklas@student.ubc.ca
- Lianne Presley lpresley@student.ubc.ca
- Thomas Worthington tworth01@mail.ubc.ca

TECHNICAL STAFF:

- Manjeet Bains (Technical Lead)
- Noushin Akhoundsadegh (Technical Support)

COMMUNICATION

We will be connecting with you in the lab and in lecture regularly. For additional help, Sree and Kabir are happy to meet with you outside class by appointment. Please reach out! If you have any concerns about course logistics or other matters you wish to discuss privately, please reach out to the course coordinator (Sree) via email at s.vappala@ubc.ca.

PRE-REQUISITES

MICB 322 (MOLECULAR MICROBIOLOGY LABORATORY)

This course is restricted to students in one of these programs: BSc with one of these specializations: Honours MBIM, OR, Major MBIM.

RECOMMENDED – MICB 212 (INTRODUCTORY MEDICAL MICROBIOLOGY AND IMMUNOLOGY)

You should be familiar with the major virology and immunology concepts discussed in MICB 212 (formerly MICB 202). We will review some key concepts before exploring these topics in more detail, but please review your notes from this course.

COURSE MATERIALS

LABS: For in-person labs, you are expected to have the following items:

- A knee-length laboratory coat.
- Eye protection.
- A bound lab notebook (see guidelines on how to keep a lab notebook posted on Canvas). Note, you are permitted to continue using the same lab notebook you used for MICB 322 as long as it is bound, and there is ample space left.
- A waterproof, permanent marker (sharpie) is appropriate for labeling materials in the lab.
- Note: It is a personal decision whether you choose to wear additional PPE such as a mask to help prevent transmission of respiratory viruses like SARS-CoV2. We support your choice to do so.

OUR CANVAS PAGE: Canvas is UBC's Learning Management System
URL www.canvas.ubc.ca - MICB 323 Section 201 – 2025W2; requires UBC CWL ID.
For Canvas help, contact the Help Desk: https://students.canvas.ubc.ca/help/

TECHNICAL REQUIREMENTS: In this course, you will need a computer/tablet with reliable internet, capable of streaming video. A smartphone will not be enough. You will need to be able to type for assignments, quizzes, and other coursework.
Your computer will also need the usual set of software, including:

- a Microsoft Office (or equivalent) suite of programs (UBC maintains licenses for many programs that allow student discounts/ downloads. See https://it.ubc.ca/software-downloads).
- at least two up-to-date web browsers (Canvas works best in Chrome; other software will be optimized for other browsers). Currently I have Safari, Firefox and Chrome on my computer.
- You may be asked to access additional data analysis software. For example, we will likely analyze data using ImageJ (FIJI - https://fiji.sc/), FlowJo, and Prism GraphPad https://www.graphpad.com/scientific-software/prism/ .

COURSE GOALS

IN MICB 323, STUDENTS WILL DEMONSTRATE LEARNING BY:

- following all safety rules and anticipating and preparing for any potential risks.
- correctly using and applying the selected laboratory techniques and tools.
- being able to perform standard calculations typically used to make solutions and make up working concentrations of reagents.
- keeping complete lab notes so that colleagues can figure out what experiments were done, why they were done, and how they were done.
- conducting experiments and working as a team to accomplish lab work.
- explaining the theory behind the experimental techniques.
- processing and analyzing raw data, and presenting data in a meaningful form.
- reporting the direct inferences from data, coming up with possible interpretations of the data, and connecting these interpretations to the research of others.
- evaluating and interpreting data from published research articles.
- demonstrating scientific writing skills and presenting the work performed in research paper format.

WHAT TO EXPECT IN MICB 323

LAB MODULE: The aim of the in-person labs is to practice technical skills at the bench that are relevant to any cellular immunology and/or virology lab. Our main focus is to introduce you to mammalian cell culture work, to build skills in working methodically and efficiently at the bench, and to build gradual independence in your lab skills. In a research lab, it is not possible to work from a perfectly pre-constructed list of instructions. Instead, researchers look up research documents, manuals, reagent information, and procedures, to design and plan their own experiments. You will be expected to look up several resources related to the experimental tasks, and to perform routine lab calculations. Practicing these skills and being resourceful to understand the lab techniques from a variety of sources will help you be prepared for labs in fourth year that require you to design your own experiments to answer research questions. Some of the techniques you will get hands-on experience with this term include:

- Bacterial transformation and plasmid isolation,
- Mammalian cell culture and aseptic technique in a biological safety cabinet,
- Cell labelling techniques and fluorescence microscopy,
- Protein isolation, western blotting,
- RNA isolation and RT-qPCR,
- ELISA
- Splenic cell isolation and Flow Cytometry
- industry-standard documentation in a lab notebook

We will use these techniques to answer aspects of the following broad RESEARCH QUESTION:

HOW DO MACROPHAGES RESPOND TO PATHOGENS?

To address this broad question, we will use an *in vitro* cell culture model of mouse macrophages stimulated with purified lipopolysaccharide (LPS) as a proxy for infection with a bacterial pathogen. You can look up the cell line we will be primarily using here: https://www.atcc.org/products/tib-67

We have the following more specific RESEARCH QUESTIONS:

1. Does short-term exposure to LPS activate the MAPK signaling pathway in J774A.1 macrophage cells?
2. Do cells change shape in response to LPS in J774A.1 mouse macrophage cells? Is the subcellular structure of the actin cytoskeleton altered?
3. Does LPS exposure lead to increased mRNA production and protein secretion of the cytokine TNF alpha in the J774A.1 mouse macrophage cells? If so, what are the kinetics of these changes?

Another important part of doing good science is being a respectful, conscientious citizen in the lab, and to work effectively on a team. Initially, everyone will work individually in the lab so everyone practices fundamental skills at the bench, and receive individualized feedback from the teaching team about your technique. Some of the later lab activities will be performed in pairs or groups of students.

LECTURE MODULE: This course is not intended to focus solely on learning various techniques. The goal is also to help you practice some fundamental aspects of the process of science. Part of "doing science" is collecting and presenting data meaningfully to an audience. During the lecture component of the course, we will spend time building skills in the following:

- engaging with and evaluating the primary literature;
- data analysis, and visualization using various software packages
- scientific writing in the form of scientific manuscripts.

Course Evaluation

Category	Assessment	Due Date*	%
BENCH and TECHNICAL SKILL BUILDING PORTFOLIO (20%)	**Personal reflections (Self-Assessments) -** General lab performance, technique at the bench and developing a science identity	See Canvas for Due dates (Start, Mid and End of semester)	4
	Preparedness and Engagement – Lab Quizzes, In-Class Work, Social Annotation Worksheet	See Canvas or below for Due dates (usually Monday before lecture)	
	Practical Assessment – Tissue culture work in the BSC	During LAB 6	8
	Documentation skills – Lab notebook	Due on Final Exam Day	8
ANALYZING and COMMUNICATING EXPERIMENTAL DATA SKILL BUILDING PORTFOLIO (80%)	**"How to read a research article" Assignment** Annotated Bibliography Worksheet and Concept Map	15-February-2026, 10 PM	15
	Formal Figure and Legend (feedback for manuscript)	22-March-2026, 10 PM	5
	Flow Cytometry Assignment	12-April-2026, 10 PM	5
	Written Scientific Manuscript (25%)		
	Peer Assessment (Pair work survey)	10-April-2026, 10 PM	5
	Manuscript	19-April-2026, 10 PM	20
	Two-Stage Final Exam**	During Exam period (April 14 – April 25), Date TBA	30 (I = 25.5%, G = 4.5%)

*A NOTE ABOUT DUE DATES

Due dates listed on Canvas and discussed in class are considered "best by" dates. We provide dates to help you stay on track and progress in an optimal way to help your learning. For example, if you prepare for your labs in advance and complete the quizzes on time, you will get the most out of the lab experience by knowing the background and understanding the procedures. If you complete the exercises on how to critically read primary papers on time, this will help you with the "how to read a research article" major assignment, and so on.

However, life happens, we get it. For any assignment, everyone is permitted a 2-day grace period without needing to contact us. If you anticipate needing more time or experiencing extraordinary challenges hindering your learning in MICB 323, please reach out to us so we can help you. For some assignments, it is extra important to submit on time, or at least within the 2-day grace period, so we can give you feedback in a timely way and coordinate how our team proceeds with the grading. These major assignments include:

- How to read a research article" Assignment
- Formal Figure and Legend – feedback for manuscript
- Written Scientific Manuscript

**Two Stage Final Exam

We have adopted a two-stage final exam to mirror the teamwork experience from the labs. The first stage shows your individual understanding, and the second lets you apply problem-solving and critical-thinking skills as a group to deepen and showcase your learning. The individual component is worth 85% and the group component is worth 15%. If you happen to score lower on the group stage than on the individual stage, the individual grade will account for the entire exam grade.

Academic Integrity and your responsibility as a UBC Student

As members of the UBC community, you are part of a large network of academics across the globe, whose primary business is ideas. How exciting! Academic integrity is the foundation on which all university endeavors are built, from the most advanced research to the smallest document created by a student for a course or club. Everything we do depends on our ability to create and share in an environment where we can trust each other to work honestly and ethically to create knowledge and build scholarship. When you registered for your first course at UBC, you agreed to uphold these values and to uphold academic integrity in all of your scholastic endeavors. We encourage all students to explore UBC resources on Academic Integrity, and to review what constitutes academic misconduct.

Breaches of academic integrity or academic misconduct have consequences. We will first meet with you to discuss the situation, but all suspected academic integrity violations will be reported to the Dean of Students as academic misconduct (violation of course rules). Consequences for academic misconduct may include a failing grade on an assignment, exam, an overall reduction in your final course grade, or a failing grade in the course, among other possibilities. Here is a non-exhaustive list of activities that are classified as academic misconduct:

- CHEATING – this includes working together on an assignment and using any additional resources to help you if you have not been instructed to do so (ex: a former student's course assignment).

- BREACHING THE RULES OF AN EXAM – this includes breaching an agreed-upon academic integrity pledge.

- PLAGIARISM – this can mean submitting work that is not your own, or not properly attributing someone's ideas with appropriate citations. Adding your name to a group assignment when you have not contributed, is also considered plagiarism. You can also plagiarize yourself! You are not permitted to submit the same work, even if it is your own, to receive credit for more than one assignment.

- ACCESSING AND SUBMITTING COURSE MATERIAL TO THIRD PARTY WEBSITES - Several commercial services (ex: Course Hero) have approached students regarding selling/posting class notes, study guides, and other course materials. Distributing the instructor's notes, exam questions and other course materials in this course is not permitted, ever. It is important for you to know that these third-party sites maintain a record of all submissions. This can have negative repercussions for your future. There have been cases where students were caught engaging in this behavior, which impacted their admission to professional school years later! *Just don't do it.* See more information below in the copyright disclaimer.

A special note about the use of Generative artificial intelligence (GenAI) tools: Gen AI tools that use large language models (LLMs) to generate written content based on a large dataset of training data, such as ChatGPT, have become widely accessible. Whilst these tools offer many possibilities to support learning, they also come with challenges and are no substitute for key goals of the course, to learn to evaluate primary scientific articles, and practice the craft of scientific writing.

You should be aware that GenAI tools can produce biased, false, or misleading content, because of the nature of the content in the training datasets. They are designed to produce the most statistically plausible text result, not necessarily the correct one. Most importantly, they cannot take responsibility for what they produce. You, on the other hand, must do this – it is a fundamental part of writing with integrity. So when engaging with these tools, you should ALWAYS verify all results from these tools against other scholarly sources, and take responsibility for their outputs. It is also important to acknowledge where GenAI tools have been used in an assessment according to what is emerging as the accepted academic convention by citing the use according to the APA style: https://apastyle.apa.org/blog/how-to-cite-chatgpt

Specific Guidelines in MICB 323:

- It is NOT acceptable to have a GenAI tool write assignments on your behalf, or to write your assignment fully in another language and use a translation tool to write it in English;
- It is acceptable to use a GenAI-driven translation tool to translate words and short phrases that you have written yourself in another language. Similarly, you may use translation tools as well as spelling and grammar editors to double-check an assignment draft that you have already written in English;
- It is acceptable to use GenAI tools to gather *preliminary* information, much in the same way one might consult Wikipedia. However, any outputs must be verified with scholarly sources.
- You are expected to cite scholarly sources, and you must cite your use of GenAI tools. We expect students to include a brief statement disclosing how it was used according to the instructions in the assignment.
- If you are unsure whether your use of GenAI is acceptable, please ask the Teaching team for guidance.

Final thoughts: The MICB 323 teaching team wants you to succeed, but we want you to do it fairly, and honestly. We feel that the use of GenAI has many positive uses, but also has the potential to stifle independent, creative thinking, and limit skill development and learning. Remember that we are members of the academic community, too, so we believe in the value of academic integrity in our scholarly work. We are also employees of UBC, so we are also expected to uphold the values (and policies) of UBC. We will hold you to the standards that we are held to, and we will investigate when there is an indication that academic misconduct has occurred.

Feel free to ask reach out to us to chat about academic integrity. Part of our job is to guide your growth as a scholar, and we would much rather you ask for clarification than unintentionally engage in academic misconduct, which has serious consequences. Sometimes students who are experiencing a lot of stress feel the only way to deal with a situation is to cheat. Please do not do this. If you reach out early, we can work something out together - s.vappala@ubc.ca.

Please visit the Academic Integrity Hub to learn more about Academic Integrity at UBC.
https://academicintegrity.ubc.ca/

COPYRIGHT DISCLAIMER:

With much of the course material posted online, there is a great temptation to download and repost course material to third party sites. This is not a good idea, or in any way fair to those that have worked so hard to create this material. Textbooks and lab instructions take years to create and keep up to date. Assignments and exams are carefully crafted and take hours of work to get right. Diagrams and figures included in lecture presentations adhere to Copyright Guidelines for UBC Faculty, Staff and Students http://copyright.ubc.ca/requirements/copyright-guidelines/ and UBC Fair Dealing Requirements for Faculty and Staff http://copyright.ubc.ca/requirements/fair- dealing/. All material uploaded to Canvas that contain diagrams and figures are used with permission of the publisher; are in the public domain; are licensed by Creative Commons; meet the permitted terms of use of UBC's library license agreements for electronic items; and/or adhere to the UBC Fair Dealing Requirements for Faculty and Staff. Access to the Canvas course site is limited to students currently registered in this course. Under no circumstance are students permitted to provide any other person with means to access this material. Distribution of this material to a third party is strictly forbidden. This includes taking pictures/ videos of copyrighted material on your own mobile device, and sharing them with others. Screenshots, photos and other recordings that you create of course material are for your own personal use, and cannot be uploaded to any websites, or shared with anyone that might do so. Note that the person who shares the work is just as liable as the person that uploads it unlawfully.

OTHER COURSE POLICIES

Academic concession

You may need to request an academic concession for medical reasons, on compassionate grounds, or in certain cases of conflicting responsibilities. Please refer to UBC's policy on Academic Concession for details.
To apply for an academic concession, please inform your instructor as soon as possible.

If you are ill

Please don't come to class or lab if you have an illness that could be transmitted to your classmates (e.g., a respiratory infection). In this class, we will be flexible so that you can prioritize your health and still succeed. If you are ill, please inform your lead lab instructor and the course coordinator (Sree). You will not lose participation marks if you miss a small number of classes due to illness. If you are ill for a long period of time, please contact Sree to discuss, and apply for an academic concession. More information about UBC's framework for preventing communicable disease is here.

Tentative Course Schedule*

***Schedule and description of topics may be subject to change.*

Dates Lab #	LECTURE Monday, 1-3pm IRC G1	LAB MODULE ACTIVITIES T/W/Th/F 1-5pm Biological Sciences 3145	NOTES & FORMAL ASSESSMENTS
Jan 5-9	Course Orientation Research project overview	No in-person labs	
LAB 1 Jan 12-16	Lab 1 overview (Aseptic work/bacterial work & Plasmids) Reading scientific papers activity Part 1	Bacterial culture aseptic technique • Bacterial transformation • Plasmid preparation	Self-Assessment 1 **Due:** Jan 12th Lab Safety Quiz Lab 1 Quiz **Due:** Jan 12th
LAB 2 Jan 19-23	Lab 2 overview (Mammalian cell culture) Reading scientific papers activity 2	Mammalian cell culture and aseptic technique • Basic tissue culture skills • Subculturing adherent cells; counting cells • Phase microscopy Record results from lab 1	*(Last day to drop without a 'W')* Lab 2 Quiz **Due:** Jan 19th
LAB 3 Jan 26-30	Lab 3 overview (Cell stimulation SDS PAGE) Reading scientific papers activity 3	Cell stimulation, Protein Extraction and quantification; Cast resolving gel for SDS-PAGE	Lab 3 Quiz **Due:** Jan 26th
LAB 4 Feb 2-6	Lab 4 Overview (SDS-PAGE, Staining)	SDS-PAGE: • Cast stacking gel, prep protein, run gel, transfer Fluorescent labeling cultured cells Formative feedback – BSC Aseptic technique	Lab 4 Quiz **Due:** Feb 2nd
LAB 5 Feb 9-13	Lab 5 overview (Western blotting, Fluorescence microscopy)	Western blotting Fluorescence microscopy Formative feedback – BSC Aseptic technique	Lab 5 Quiz **Due:** Feb 9th Self-Assessment 2 **Due:** Feb 15th Annotated Bibliography Worksheet and Concept Map **Due:** Feb 15th
Feb 16-20	FAMILY DAY/READING WEEK		
LAB 6 Feb 23-27	Making figures and writing figure legends Data analysis workshops: Image J – Analyzing images Image J – Western blot quantification	Practical Assessment – Tissue culture work in the BSC	IN LAB Formal assessment – BSC Aseptic technique
LAB 7 Mar 2-6	Lab 7 overview (ELISA)	ELISA	Lab 7 Quiz **Due:** Mar 6th
LAB 8 Mar 9-13	Lab 8 Overview (RNA Extraction and qPCR) Data analysis Workshop - ELISA	RNA Extraction and qPCR	
LAB 9 Mar 16-20	Lab 9 overview (Splenic cell isolation and surface antigen labeling) Lab 10 overview (Flow cytometry) Data analysis Workshop: qPCR Data analysis and formal figure support	Splenic cell isolation and surface antigen labeling	Formal Figure and figure legend (individual) **Due:** Mar 22nd
LAB 10 Mar 23-27	Scientific Writing Storyboarding our manuscript - Recap of lab activities	Field trip to the LSI Flow Cytometry Facility Flow cytometry and Data Analysis workshop – (FlowJo)	
NO LAB Mar 30-Apr 3	Flow Cytometry debrief Flow Cytometry Assignment support	FLEX WEEK – NO LABS In lab – data analysis and writing support	

NO LAB Apr 6-10	Preparing your manuscript Peer Review Session	FLEX WEEK – NO LABS In lab – data analysis and writing support	Peer Assessment (Pair work survey) **Due:** April 10th Flow Cytometry Assignment **Due:** April 12th Manuscript (in pairs) **Due:** April 19th (date may change depending on final exam date)
Final Exam period Apr 14-25	Final exam date TBA (exam schedule released Mid-Feb)		**Due on final exam day:** Lab Notebook Self-Assessment 3